

Lab 5: Collaboration and VOPC Diagrams

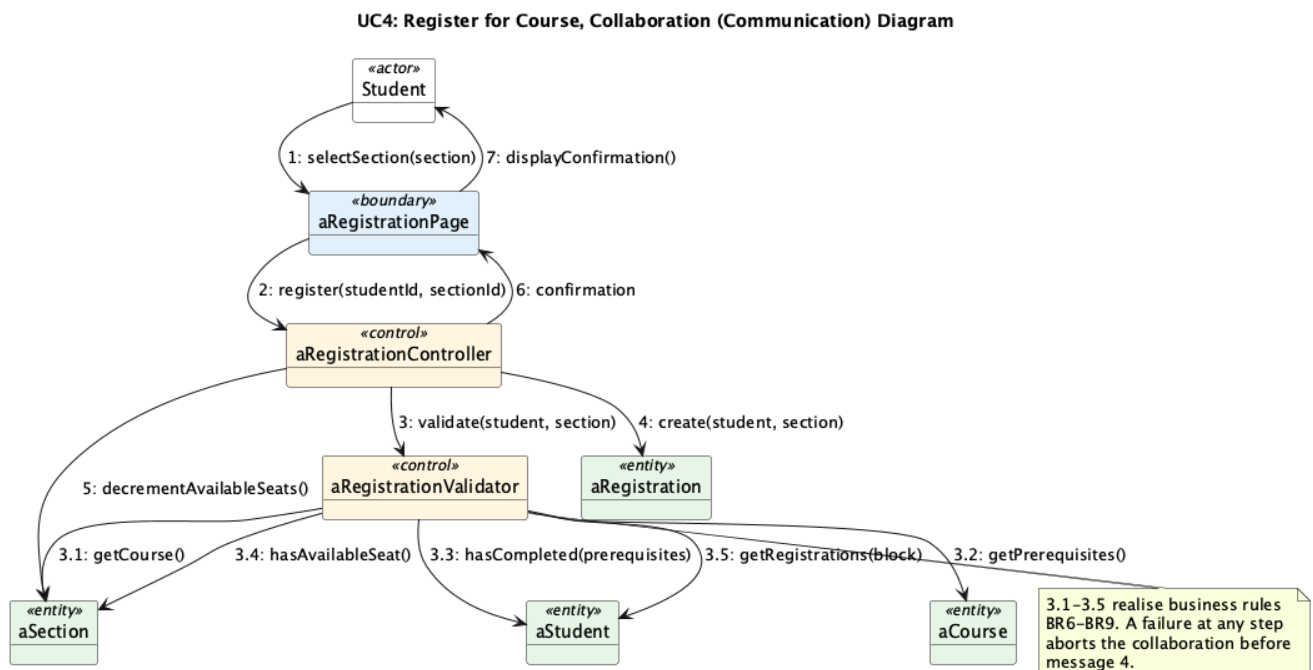
Project: eRegistrar

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For each of the two major use cases, this lab pairs a **collaboration (communication) diagram** (the same interaction as Lab 4, but organised around the links between objects, with hierarchical message numbering) with a **VOPC (View of Participating Classes)** diagram showing the analysis classes with their attributes, operations and relationships.

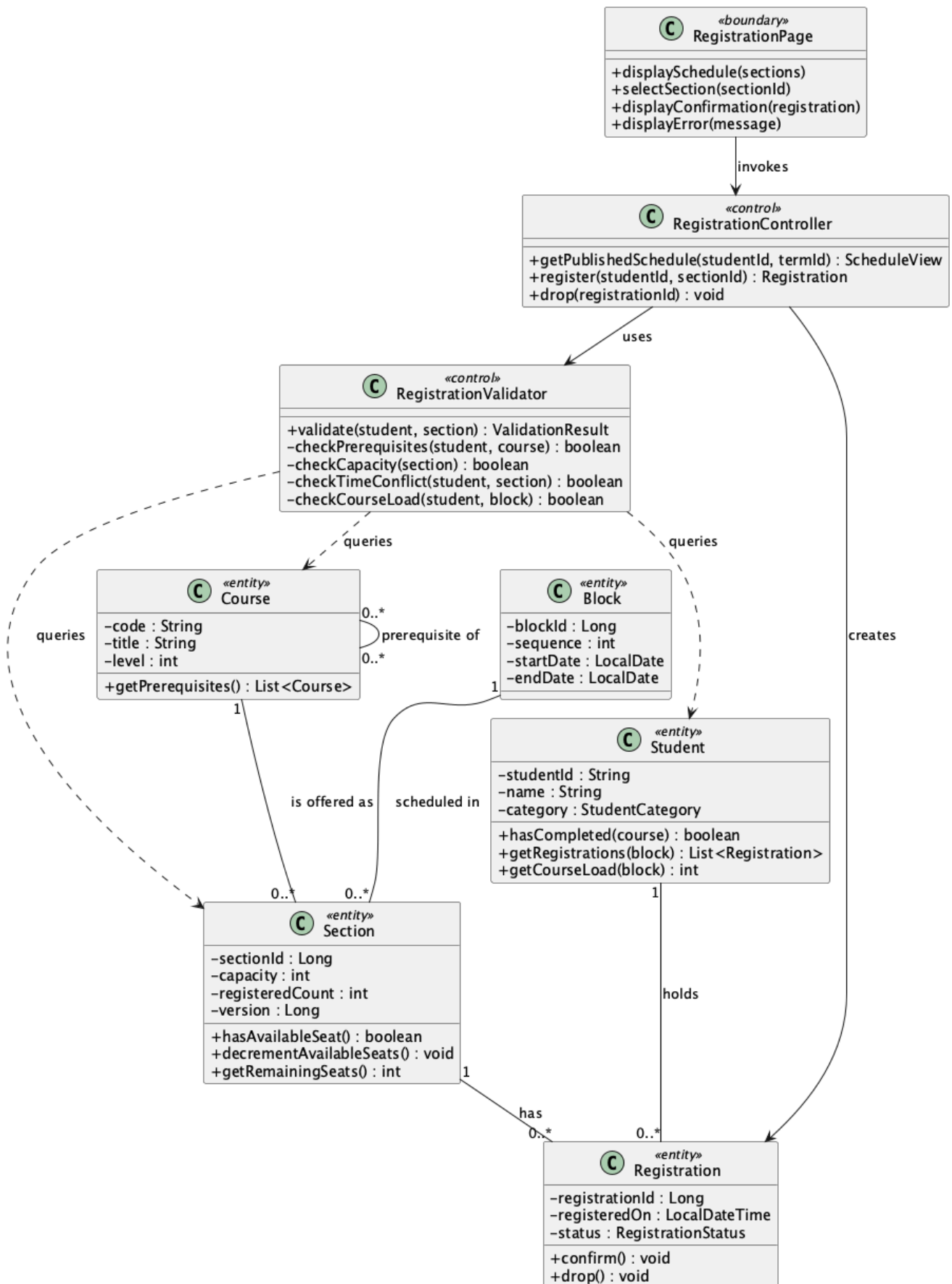
UC4: Register for Course

Collaboration diagram



Messages 3.1 through 3.5 are the validation sub-messages sent by `aRegistrationValidator` while handling message 3. If any of them fails, the collaboration stops before message 4: no registration is created and no seat is consumed.

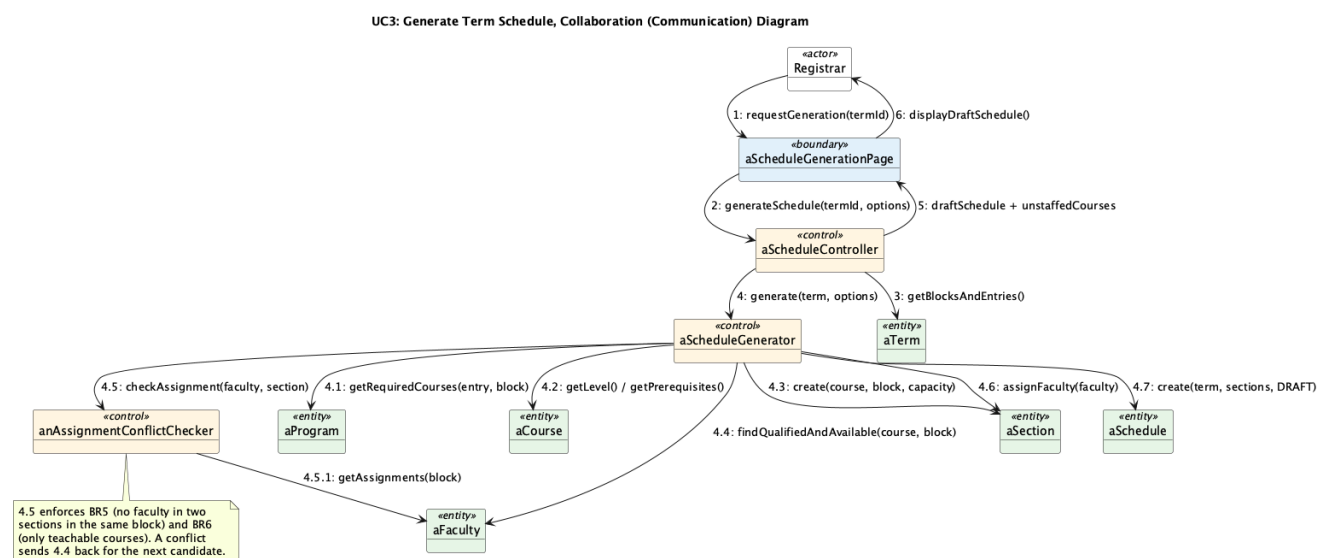
UC4: Register for Course, VOPC (View of Participating Classes)



Class	Stereotype	Responsibility
RegistrationPage	boundary	Presents the schedule, collects the selection, shows the result.
RegistrationController	control	Coordinates the use case and owns the transaction.
RegistrationValidator	control	Prerequisites, capacity, time conflict, course load (BR6 to BR9).
Student	entity	Category, completed courses, current registrations.
Section	entity	Capacity, registered count, and <code>version</code> for optimistic locking.
Course	entity	Level and prerequisites (the reflexive "prerequisite of" association).
Registration	entity	The student-to-section association.
Block	entity	The 8-week period a section is scheduled in.

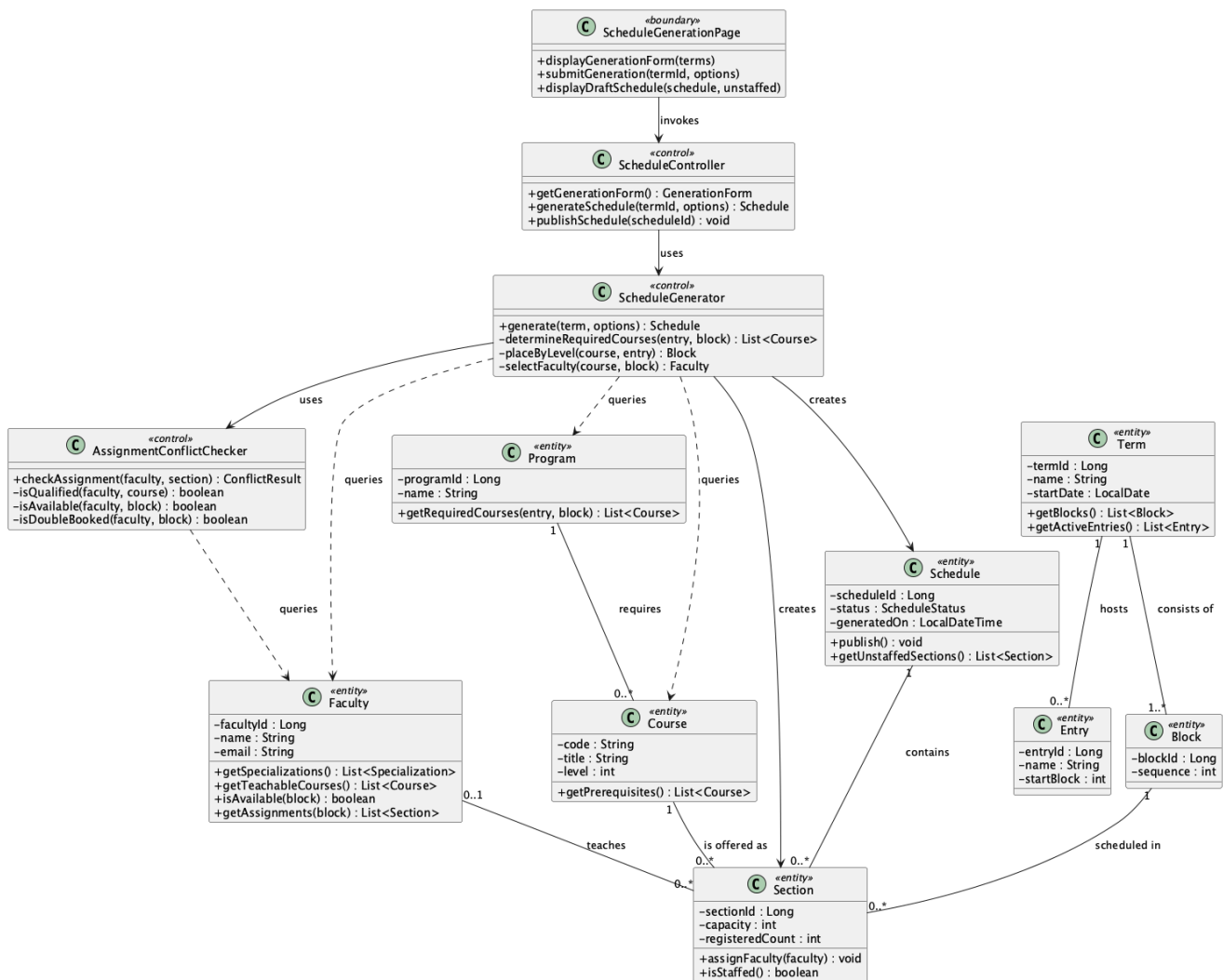
UC3: Generate Term Schedule

Collaboration diagram



Messages 4.1 through 4.7 are the generation steps performed by `aScheduleGenerator`. Message 4.5.1 is the conflict checker's query back to `aFaculty`, which enforces BR3 and BR4.

UC3: Generate Term Schedule, VOPC (View of Participating Classes)



Class	Stereotype	Responsibility
ScheduleGenerationPage	boundary	Generation form and draft-schedule display.
ScheduleController	control	Coordinates generation.
ScheduleGenerator	control	Required courses, block placement, section creation, faculty selection.
AssignmentConflictChecker	control	Qualification, availability and double-booking checks.
Term, Entry, Block, Program, Course, Faculty, Section, Schedule	entity	The scheduling data.

How these map to the design

Control classes become business-layer services, entity classes become the domain model and JPA entities, and boundary classes become Spring MVC controllers with their Thymeleaf views. That's the

layering set out in the architecture document (Lab 3), and it's realised in the Lab 7 eRegistrar application.

Re-rendering

```
java -jar tools/plantuml.jar -tpng Lab5_Collaboration_VOPC/diagrams/*.puml
```